

Lemma:  $n, r \in \mathbb{Z}_{\geq 1}$  且  $r \leq n$ , 则有:  $\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}$

Proof: 右边  $= \frac{n}{r} \cdot \binom{n-1}{r-1} = \frac{n}{r} \cdot \frac{(n-1)!}{(r-1)!(n-r)!} = \frac{n!}{r!(n-r)!}$

$= \binom{n}{r} = \text{左边} \quad \square$

Lemma:  $G$  是有限群,  $p$  是素数, 则对  $\forall P \in \text{Syl}_p(G)$ , 有:

$$\{xPx^{-1} \mid x \in G\} \subseteq \text{Syl}_p(G).$$

Proof:  $\because G$  是有限群,  $p$  是素数  $\therefore |G| = p^m \cdot k$ , 其中  $m \in \mathbb{Z}_{\geq 0}, k \in \mathbb{Z}_{\geq 1}$ ,

$p \nmid k \therefore p^m \parallel |G|$ . 对  $\forall P \in \text{Syl}_p(G)$ , 有:

任取集合  $\{xPx^{-1} \mid x \in G\}$  中的一个元素  $xPx^{-1}$  (其中  $x \in G$ )

$\because G$  是有限群,  $P \leq G, x \in G \therefore xPx^{-1} \leq G$  且  $|xPx^{-1}| = |P|$

$\because P \in \text{Syl}_p(G) \therefore |P| = p^m \therefore |xPx^{-1}| = p^m$

$\therefore xPx^{-1} \in \text{Syl}_p(G) \therefore \{xPx^{-1} \mid x \in G\} \subseteq \text{Syl}_p(G) \quad \square$

Lemma:  $G$  是有限群,  $p$  是素数, 则对  $\forall P \in \text{Syl}_p(G)$ , 有:

$$p \nmid (G:P) \text{ 且 } (G:P) \in \mathbb{Z}_{\geq 1}$$

Proof:  $\because P \in \text{Syl}_p(G) \therefore P \leq G \because G$  是有限群,  $p$  是素数

$$\therefore |G| = p^m \cdot k, \text{ 其中 } m \in \mathbb{Z}_{\geq 0}, k \in \mathbb{Z}_{\geq 1}, p \nmid k, p^m \parallel |G|.$$

$$\therefore |P| = p^m \quad \because G \text{ 是有限群, } P \leq G \quad \therefore |G| = |P| \cdot (G:P)$$

$$\therefore (G:P) = \frac{|G|}{|P|} = \frac{p^m \cdot k}{p^m} = k \in \mathbb{Z}_+ \quad \therefore p \nmid k \quad \therefore p \nmid (G:P) \quad \square$$

Lemma: 群  $G$  作用在非空集合  $X$  上. 记这个群作用为

$$\alpha: G \times X \rightarrow X. \quad H \leq G. \quad \text{则 } b: H \times X \rightarrow X$$

$$(g, x) \mapsto \alpha(g, x)$$

是子群  $H$  在  $X$  上的群作用.

Proof: 对  $\forall (g, x) \in H \times X$ , 有:  $g \in H$  且  $x \in X$ .

$$\therefore g \in G \text{ 且 } x \in X \quad \therefore b(g, x) = \alpha(g, x) \in X.$$

对  $\forall (g_1, x_1), (g_2, x_2) \in H \times X$ , 若  $(g_1, x_1) = (g_2, x_2)$ , 则

$$g_1 = g_2 \text{ 且 } x_1 = x_2. \quad \therefore g_1 = g_2 \in H \leq G \text{ 且 } x_1 = x_2 \in X$$

$$\therefore b(g_1, x_1) = \alpha(g_1, x_1) = \alpha(g_2, x_2) = b(g_2, x_2)$$

$\therefore b: H \times X \rightarrow X$  是良定义的映射.

$$\text{对 } \forall x \in X, \text{ 有: } b(1_G, x) = \alpha(1_G, x) = x.$$

$$\text{对 } \forall g_1, g_2 \in H, \forall x \in X, \text{ 有: } g_1 g_2 \in H$$

$$\therefore b(g_1, b(g_2, x)) = b(g_1, \alpha(g_2, x)) = \alpha(g_1, \alpha(g_2, x))$$

$$= \alpha(g_1 g_2, x) = b(g_1 g_2, x)$$

$\therefore b: H \times X \rightarrow X$  是群  $H$  在非空集合  $X$  上的群作用  $\square$ .

Lemma:  $G$  是群,  $P \leq G$ , 定义集合  $X = \{xPx^{-1} \mid x \in G\}$

$H \leq G$ , 定义群作用  $\alpha: H \times X \longrightarrow X$

$$(g, xPx^{-1}) \mapsto (gx)P(x^{-1}g^{-1})$$

则有:  $\alpha$  是群  $H$  在非空集合  $X$  上的群作用.

Proof: 对  $\forall (g, xPx^{-1}) \in H \times X$ , 有:  $g \in H$  且  $xPx^{-1} \in X$

且  $x \in G$ .  $\therefore gx \in G$  且  $\alpha(g, xPx^{-1}) = (gx)P(x^{-1}g^{-1})$

$= (gx)P(gx)^{-1} \in X$  显然  $1_G P 1_G^{-1} = P \in X \therefore X \neq \emptyset$ .

对  $\forall (g_1, x_1Px_1^{-1}), (g_2, x_2Px_2^{-1}) \in H \times X$ . 若

$(g_1, x_1Px_1^{-1}) = (g_2, x_2Px_2^{-1})$ , 则有  $g_1 = g_2 \in H$  且

$x_1Px_1^{-1} = x_2Px_2^{-1} \in X$  (其中  $x_1, x_2 \in G$ ).

$\therefore \alpha(g_1, x_1Px_1^{-1}) = (g_1x_1)P(x_1^{-1}g_1^{-1}) = (g_1x_1)P(g_1x_1)^{-1}$

$\alpha(g_2, x_2Px_2^{-1}) = (g_2x_2)P(x_2^{-1}g_2^{-1}) = (g_2x_2)P(g_2x_2)^{-1}$

任取  $(g_1x_1)P(x_1^{-1}g_1^{-1})$  中的一元:  $g_1x_1\alpha x_1^{-1}g_1^{-1}$  ( $\alpha \in P$ )

$\therefore x_1\alpha x_1^{-1} \in x_1Px_1^{-1} = x_2Px_2^{-1} \therefore \exists \beta \in P, \text{ s.t. }$

$$x_1\alpha x_1^{-1} = x_2\beta x_2^{-1}$$

$$\because g_1 x_1 \alpha x_1^{-1} g_1^{-1} = g_1 x_2 \beta x_2^{-1} g_1^{-1} = g_2 x_2 \beta x_2^{-1} g_2^{-1} \in (g_2 x_2) P (x_2^{-1} g_2^{-1})$$

$$\therefore (g_1 x_1) P (x_1^{-1} g_1^{-1}) \subseteq (g_2 x_2) P (x_2^{-1} g_2^{-1})$$

$$\text{同理可证 } (g_2 x_2) P (x_2^{-1} g_2^{-1}) \subseteq (g_1 x_1) P (x_1^{-1} g_1^{-1})$$

$$\therefore (g_1 x_1) P (x_1^{-1} g_1^{-1}) = (g_2 x_2) P (x_2^{-1} g_2^{-1})$$

$$\therefore \alpha(g_1, x_1 P x_1^{-1}) = \alpha(g_2, x_2 P x_2^{-1}) \quad \therefore \alpha \text{ 是良定义的映射.}$$

对  $\forall x P x^{-1} \in X$  (其中  $x \in G$ ), 有:

$$\alpha(1_G, x P x^{-1}) = (1_G x) P (x^{-1} 1_G^{-1}) = x P x^{-1}$$

对  $\forall g_1, g_2 \in H, \forall x P x^{-1} \in X$  (其中  $x \in G$ ), 有:

$$\because g_1, g_2 \in H \quad \therefore g_1 g_2 \in H$$

$$\therefore \alpha(g_1, \alpha(g_2, x P x^{-1})) = \alpha(g_1, (g_2 x) P (x^{-1} g_2^{-1}))$$

$$\cancel{\alpha(g_1, (g_2 x) P (x^{-1} g_2^{-1}))} = \alpha(g_1, (g_2 x) P (g_2 x)^{-1})$$

$$= (g_1 (g_2 x)) P ((g_2 x)^{-1} g_1^{-1}) = ((g_1 g_2) x) P ((g_1 g_2) x)^{-1}$$

$$= ((g_1 g_2) x) P (x^{-1} (g_1 g_2)^{-1}) = \alpha(g_1 g_2, x P x^{-1})$$

$\therefore \alpha$  是群  $H$  在非空集合  $X$  上的良定义的群作用.  $\square$

Lemma:  $p$  是素数,  $a \in \mathbb{Z}_{\geq 1}$ ,  $m, k \in \mathbb{Z}_{\geq 0}$ ,  $p \nmid a$ ,  $k \leq m$ .

则有:

$$\textcircled{1} p^{m-k} \parallel \binom{p^m \cdot a}{p^k} \quad \textcircled{2} \frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} \equiv a \pmod{p}$$

$$\text{Proof: } \because m \in \mathbb{Z}_{\geq 0} \quad \therefore p^m \in \mathbb{Z}_{\geq 1} \quad \because a \in \mathbb{Z}_{\geq 1} \quad \therefore p^m \cdot a \in \mathbb{Z}_{\geq 1}$$

$$\because k \in \mathbb{Z}_{\geq 0} \quad \therefore p^k \in \mathbb{Z}_{\geq 1} \quad \because k \leq m \quad \therefore p^k \leq p^m$$

$$\because p^m \geq p^k \geq 1, \quad a \geq 1 > 0 \quad \therefore p^m \cdot a \geq p^k \geq 1 \quad \therefore \binom{p^m \cdot a}{p^k} \in \mathbb{Z}_{\geq 1}$$

$$\because m-k \in \mathbb{Z}_{\geq 0} \quad \therefore p^{m-k} \in \mathbb{Z}_{\geq 1}$$

$$\text{当 } k=0 \text{ 时, } p^k = p^0 = 1, \quad \binom{p^m \cdot a}{p^k} = \binom{p^m \cdot a}{1} = p^m \cdot a$$

$$p^{m-k} = p^{m-0} = p^m. \quad \because p \nmid a \quad \therefore p^m \parallel p^m \cdot a \quad \therefore p^{m-k} \parallel \binom{p^m \cdot a}{p^k} \quad \textcircled{1} \text{得证.}$$

$$\frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} = \frac{1}{p^m} p^m \cdot a = a \equiv a \pmod{p} \quad \textcircled{2} \text{得证.}$$

$$\text{下设 } k \in \mathbb{Z}_{\geq 1}. \quad \therefore p^k \geq p^1 = p \geq 2 > 1 \quad \therefore p^k \in \mathbb{Z}_{\geq 2} \quad \therefore p^k - 1 \in \mathbb{Z}_{\geq 1}$$

$$\because p^m \cdot a \in \mathbb{Z}_{\geq 1}, \quad p^k \in \mathbb{Z}_{\geq 1}, \quad p^k \leq p^m \cdot a$$

$$\therefore \binom{p^m \cdot a}{p^k} = \frac{p^m \cdot a}{p^k} \binom{p^m \cdot a - 1}{p^k - 1} \quad \text{记 } B = \binom{p^m \cdot a - 1}{p^k - 1}$$

$$\because 1 \leq p^k \leq p^{m \cdot a} \quad \therefore 0 \leq p^k - 1 \leq p^{m \cdot a} - 1$$

$$\because p^{m \cdot a} - 1 \in \mathbb{Z}_{\geq 0}, \quad p^k - 1 \in \mathbb{Z}_{\geq 0} \quad \therefore \binom{p^{m \cdot a} - 1}{p^k - 1} \in \mathbb{Z}_{\geq 1}$$

$$\therefore B \in \mathbb{Z}_{\geq 1} \quad \therefore \binom{p^{m \cdot a}}{p^k} = p^{m-k \cdot a} \cdot B$$

$$\therefore B = \binom{p^{m \cdot a} - 1}{p^k - 1} = \frac{(p^{m \cdot a} - 1)(p^{m \cdot a} - 2) \cdots (p^{m \cdot a} - (p^k - 1))}{(p^k - 1)(p^k - 2) \cdots (p^k - (p^k - 1))}$$

$$= \frac{(p^{m \cdot a} - 1)(p^{m \cdot a} - 2) \cdots (p^{m \cdot a} - (p^k - 1))}{1 \times 2 \times \cdots \times (p^k - 1)} = \prod_{i=1}^{p^k-1} \frac{p^{m \cdot a} - i}{i}$$

对  $\forall i = 1, 2, \dots, p^k - 1$ , 定义  $r_i = v_p(i)$

(因为  $p$  是素数,  $i$  是正整数  $\therefore$  存在唯一的  $v_p(i) \in \mathbb{Z}_{\geq 0}$ , s.t.

$$p^{v_p(i)} \parallel i \quad \therefore p^{r_i} \parallel i \text{ 且 } r_i \in \mathbb{Z}_{\geq 0})$$

$$\text{定义 } c_i = \frac{i}{p^{r_i}} \quad \therefore p^{r_i} \parallel i \quad \therefore p^{r_i} \mid i \text{ 且 } p^{r_i+1} \nmid i$$

$$\because r_i \in \mathbb{Z}_{\geq 0} \quad \therefore p^{r_i} \in \mathbb{Z}_{\geq 1} \quad \because i \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{i}{p^{r_i}} \in \mathbb{Z}_{\geq 1} \quad \therefore c_i \in \mathbb{Z}_{\geq 1}$$

$$\text{假设 } p \mid c_i. \text{ 则 } \frac{c_i}{p} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{i}{p^{r_i+1}} \in \mathbb{Z}_{\geq 1} \quad \therefore p^{r_i+1} \mid i \text{ 矛盾.}$$

$$\therefore p \nmid c_i \quad \therefore c_i \in \mathbb{Z}_{\geq 1} \text{ 且 } p \nmid c_i \quad \therefore c_i = \frac{i}{p^{r_i}}, \quad p^{r_i} \in \mathbb{Z}_{\geq 1}$$

$$\therefore i = p^{r_i} \cdot c_i, \quad r_i \in \mathbb{Z}_{\geq 0}, \quad c_i \in \mathbb{Z}_{\geq 1}, \quad p \nmid c_i$$

$$\because C_i \geq 1, \quad p^{r_i} > 0 \quad \therefore p^{r_i} \cdot C_i \geq p^{r_i} \quad \therefore p^{r_i} \leq i < p^k$$

$$\therefore r_i < k \quad \therefore r_i < k \leq m \quad \therefore r_i < m \quad \therefore m - r_i > 0$$

$$\therefore m - r_i \in \mathbb{Z}_{\geq 1} \quad \therefore p^m \cdot a - i = p^m \cdot a - p^{r_i} \cdot C_i = p^{r_i} (p^{m-r_i} \cdot a - C_i)$$

$$\therefore \frac{p^m \cdot a - i}{i} = \frac{p^{r_i} (p^{m-r_i} \cdot a - C_i)}{p^{r_i} \cdot C_i} = \frac{p^{m-r_i} \cdot a - C_i}{C_i}$$

$$\therefore B = \prod_{i=1}^{p^k-1} \frac{p^m \cdot a - i}{i} = \prod_{i=1}^{p^k-1} \frac{p^{m-r_i} \cdot a - C_i}{C_i}$$

$$= \frac{\prod_{i=1}^{p^k-1} (p^{m-r_i} \cdot a - C_i)}{\prod_{i=1}^{p^k-1} C_i}$$

$$\text{定义 } U = \prod_{i=1}^{p^k-1} (p^{m-r_i} \cdot a - C_i)$$

$$\text{定义 } V = \prod_{i=1}^{p^k-1} C_i \quad \therefore B = \frac{U}{V}$$

$$\therefore \text{对 } \forall i=1, 2, \dots, p^k-1, \text{ 有: } C_i \in \mathbb{Z}_{\geq 1} \quad \therefore \prod_{i=1}^{p^k-1} C_i \in \mathbb{Z}_{\geq 1}$$

$$\therefore V \in \mathbb{Z}_{\geq 1} \quad \text{假设 } p \mid V, \text{ 则 } p \mid C_1 C_2 \cdots C_{p^k-1}$$

$$\therefore p \mid C_1 \text{ 或 } p \mid C_2 \text{ 或 } \cdots \text{ 或 } p \mid C_{p^k-1} \text{ 矛盾. } \therefore p \nmid V \quad \therefore V \in \mathbb{Z}_{\geq 1} \text{ 且 } p \nmid V$$

$$\equiv \text{对 } \forall i=1, 2, \dots, p^k-1, \quad \therefore m - r_i \in \mathbb{Z}_{\geq 1} \quad \therefore p^{m-r_i} \in \mathbb{Z}_{\geq 1}$$

$$\therefore a \in \mathbb{Z}_{\geq 1} \quad \therefore p^{m-r_i} \cdot a \in \mathbb{Z}_{\geq 1} \quad \therefore C_i \in \mathbb{Z}_{\geq 1} \quad \therefore p^{m-r_i} \cdot a - C_i \in \mathbb{Z}$$

$$\therefore i = p^{r_i} \cdot C_i, \quad i < p^k \quad \therefore p^{r_i} \cdot C_i < p^k \quad \therefore p^{r_i} > 0 \quad \therefore C_i < p^{k-r_i}$$

$$\therefore k \leq m \quad \therefore k - r_i \leq m - r_i \quad \therefore C_i < p^{k-r_i} \leq p^{m-r_i}$$

$$\because a \geq 1, \quad p^{m-r_i} > 0 \quad \therefore p^{m-r_i} \cdot a \geq p^{m-r_i}$$

$$\therefore c_i < p^{k-r_i} \leq p^{m-r_i} \leq p^{m-r_i} \cdot a \quad \therefore p^{m-r_i} \cdot a - c_i > 0$$

$$\therefore p^{m-r_i} \cdot a - c_i \in \mathbb{Z}_{\geq 1} \quad \therefore \prod_{i=1}^{p^k-1} (p^{m-r_i} \cdot a - c_i) \in \mathbb{Z}_{\geq 1}$$

$$\therefore U \in \mathbb{Z}_{\geq 1} \quad \therefore B = \frac{U}{V}, \quad B \in \mathbb{Z}_{\geq 1}, \quad U \in \mathbb{Z}_{\geq 1}, \quad V \in \mathbb{Z}_{\geq 1}$$

$$\therefore U = BV$$

$$\forall i=1, 2, \dots, p^k-1, \quad \therefore (p^{m-r_i} \cdot a - c_i) - (-c_i) = p^{m-r_i} \cdot a$$

$$p \mid p^{m-r_i} \cdot a \quad \therefore p^{m-r_i} \cdot a - c_i \equiv -c_i \pmod{p}$$

$$\therefore \prod_{i=1}^{p^k-1} (p^{m-r_i} \cdot a - c_i) \equiv \prod_{i=1}^{p^k-1} (-c_i) \pmod{p}$$

$$\therefore U \equiv (-1)^{p^k-1} \prod_{i=1}^{p^k-1} c_i \pmod{p} \quad \therefore U \equiv (-1)^{p^k-1} V \pmod{p}$$

$$\text{如果 } p=2, \text{ 则 } \therefore k \in \mathbb{Z}_{\geq 1} \quad \therefore p^k = 2^k \text{ 是偶数} \quad \therefore p^k-1 \text{ 是奇数}$$

$$\therefore (-1)^{p^k-1} = -1 \quad \therefore -1 \equiv 1 \pmod{2} \quad \therefore (-1)^{p^k-1} \equiv 1 \pmod{p}$$

$$\text{如果 } p \text{ 是奇素数, 则 } \therefore k \in \mathbb{Z}_{\geq 1} \quad \therefore p^k \text{ 是奇数} \quad \therefore p^k-1 \text{ 是偶数}$$

$$\therefore (-1)^{p^k-1} = 1 \quad \therefore 1 \equiv 1 \pmod{p} \quad \therefore (-1)^{p^k-1} \equiv 1 \pmod{p}$$

$$\therefore \text{对任意的素数 } p, \text{ 都有 } (-1)^{p^k-1} \equiv 1 \pmod{p} \quad \therefore V \in \mathbb{Z}_{\geq 1}$$

$$\therefore (-1)^{p^k-1} V \equiv V \pmod{p} \quad \therefore U \equiv V \pmod{p}$$

$$\therefore BV \equiv V \pmod{p} \quad \therefore p \mid BV - V \quad \therefore p \mid (B-1)V$$

$$\therefore p \nmid V \quad \therefore \gcd(p, V) = 1$$

$$\therefore p \in \mathbb{Z}_{\geq 2}, V \in \mathbb{Z}_{\geq 1}, B-1 \in \mathbb{Z}_{\geq 0}, p \neq 0, \gcd(p, V) = 1$$

$$p \mid V(B-1) \quad \therefore p \mid B-1 \quad \therefore B \equiv 1 \pmod{p} \quad \therefore a \in \mathbb{Z}_{\geq 1}$$

$$\therefore aB \equiv a \pmod{p} \quad \therefore \frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} = a \cdot B \in \mathbb{Z}_{\geq 1}$$

$$\therefore \frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} \equiv a \pmod{p} \quad \textcircled{2} \text{得证.}$$

$$\therefore \frac{\binom{p^m \cdot a}{p^k}}{p^{m-k}} \in \mathbb{Z}_{\geq 1} \quad \therefore p^{m-k} \mid \binom{p^m \cdot a}{p^k}$$

$$\text{假设 } p^{m-k+1} \mid \binom{p^m \cdot a}{p^k}, \text{ 则 } \therefore \binom{p^m \cdot a}{p^k} \in \mathbb{Z}_{\geq 1}, p^{m-k+1} \in \mathbb{Z}_{\geq 1}$$

$$\therefore \frac{\binom{p^m \cdot a}{p^k}}{p^{m-k+1}} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{\binom{p^m \cdot a}{p^k}}{p^{m-k} \cdot p} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{\frac{\binom{p^m \cdot a}{p^k}}{p^{m-k}}}{p} \in \mathbb{Z}_{\geq 1}$$

$$\therefore p \mid \frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} \quad \therefore \frac{1}{p^{m-k}} \binom{p^m \cdot a}{p^k} \equiv a \pmod{p} \quad \therefore p \mid a$$

$$\text{矛盾.} \quad p^{m-k+1} \nmid \binom{p^m \cdot a}{p^k} \quad \therefore p^{m-k} \parallel \binom{p^m \cdot a}{p^k} \quad \textcircled{1} \text{得证.} \quad \square$$

定理 (Sylow 第一定理)  $G$  是有限群,  $p$  是素数,  $k \in \mathbb{Z}_{\geq 0}$ .

$p^k \mid |G|$ , 则  $G$  中一定有为阶为  $p^k$  的子群.

Proof: 当  $k=0$  时,  $p^k = p^0 = 1$ .  $G$  中一定有为阶为  $p^k = 1$  的子群  $\{1_G\}$ .  $\therefore$  当  $k=0$  时定理得证. 下设  $k \in \mathbb{Z}_{\geq 1}$ .  $\because G$  是有限群

$\therefore |G| \in \mathbb{Z}_{\geq 1}$ .  $\because k \in \mathbb{Z}_{\geq 1}$  且  $p^k \mid |G|$   $\therefore$  设  $|G| = p^m \cdot a$  其中

$m \in \mathbb{Z}_{\geq 1}$  且  $k \leq m$ ,  $a \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid a$   $\because k \geq 1 \therefore p^k \geq p^1 = p \geq 2 \therefore p^k \in \mathbb{Z}_{\geq 2}$

定义集合  $Y = \{E \subseteq G \mid |E| = p^k\}$

$\therefore |Y| = \binom{p^m \cdot a}{p^k} \therefore p^{m-k} \mid |Y|$  (其中  $m-k \in \mathbb{Z}_{\geq 0}$ ),  $|Y| \in \mathbb{Z}_{\geq 1}$

$\therefore p^{m-k} \mid |Y|$  且  $p^{m-k+1} \nmid |Y|$

定义群  $G$  在集合  $Y$  上的群作用  $\alpha: G \times Y \longrightarrow Y$   
 $(g, E) \longmapsto gE$ .

对  $\forall (g, E) \in G \times Y$ , 其中  $g \in G$  且  $E \in Y$   $\therefore g \in G$  且  $E \subseteq G$  且  $|E| = p^k$   $\therefore \alpha(g, E) = gE \subseteq G$  且  $|\alpha(g, E)| = |gE| = |E| = p^k$   
 $\therefore \alpha(g, E) \in Y$

对  $\forall (g_1, E_1), (g_2, E_2) \in G \times Y$ , 若  $(g_1, E_1) = (g_2, E_2)$ , 则  
 $g_1 = g_2$  且  $E_1 = E_2$ .  $\therefore \alpha(g_1, E_1) = g_1 E_1 \stackrel{\downarrow}{=} g_2 E_2 = \alpha(g_2, E_2)$   
双方向包含显然

$\therefore \alpha: G \times Y \rightarrow Y$  是良定义的映射.

$$\text{对 } \forall E \in Y, \text{ 有: } E \subseteq G \quad \therefore \alpha(1_G, E) = 1_G E = \{1_G x \mid x \in E\} \\ = \{x \mid x \in E\} = E.$$

$$\text{对 } \forall g_1, g_2 \in G, \forall E \in Y, \text{ 有: } E \subseteq G$$

$$\therefore \alpha(g_1, \alpha(g_2, E)) = \alpha(g_1, g_2 E) = g_1(g_2 E) = (g_1 g_2) E \\ = \alpha(g_1 g_2, E) \quad \therefore \alpha: G \times Y \rightarrow Y \text{ 是群 } G \text{ 在集合 } Y \text{ 上的群作用,}$$

且  $Y$  是非空有限集合. 设所有的  $G$ -轨道为  $GE_1, GE_2, \dots, GE_\lambda$

(其中  $\lambda \in \mathbb{Z}_{\geq 1}, E_1, E_2, \dots, E_\lambda \in Y$ )

$$\therefore Y = \bigsqcup_{i=1}^{\lambda} GE_i, \text{ 且对 } \forall i=1, 2, \dots, \lambda, \text{ 都有 } GE_i \neq \emptyset$$

$$\therefore |Y| = \sum_{i=1}^{\lambda} (G : \text{Stab}_G(E_i))$$

$$\text{对 } \forall i=1, 2, \dots, \lambda, (G : \text{Stab}_G(E_i)) = |G / \text{Stab}_G(E_i)| = |GE_i| \in \mathbb{Z}_{\geq 1}$$

$$\therefore p^{m-k+1} \nmid |Y| \quad \therefore \exists \eta \in \{1, 2, \dots, \lambda\}, \text{ s.t. } p^{m-k+1} \nmid (G : \text{Stab}_G(E_\eta))$$

$$\therefore E_\eta \in Y \quad \therefore \text{Stab}_G(E_\eta) \leq G \quad \therefore |\text{Stab}_G(E_\eta)| \in \mathbb{Z}_{\geq 1}$$

$$\therefore |G| \in \mathbb{Z}_{\geq 1}, (G : \text{Stab}_G(E_\eta)) \in \mathbb{Z}_{\geq 1}$$

$$\therefore |G| = |\text{Stab}_G(E_\eta)| \cdot (G : \text{Stab}_G(E_\eta))$$

$$\text{设 } (G : \text{Stab}_G(E_\eta)) = p^r \cdot b \text{ (其中 } r \in \mathbb{Z}_{\geq 0}, b \in \mathbb{Z}_{\geq 1}, p \nmid b)$$

$$\therefore p^m \cdot a = |Stab_G(E_1)| \cdot p^r \cdot b \quad \therefore p^{m-k+1} \nmid p^r \cdot b \quad \therefore r < m-k+1$$

$$\therefore r \in \mathbb{Z}_{\geq 0}, m \in \mathbb{Z}_{\geq 1}, k \in \mathbb{Z}_{\geq 1} \quad \therefore r \leq m-k \quad \therefore k \leq m-r$$

$$\therefore p^{m-r} \cdot a = |Stab_G(E_1)| \cdot b \quad \therefore p^k \cdot p^{m-r-k} \cdot a = |Stab_G(E_1)| \cdot b$$

$$\therefore p^k \mid |Stab_G(E_1)| \cdot b \quad \therefore p \nmid b \quad \therefore p^k \nmid b \quad \therefore p^k \mid |Stab_G(E_1)|$$

$$\therefore p^k \leq |Stab_G(E_1)|$$

$$\therefore E_1 \in Y \quad \therefore E_1 \subseteq G \text{ 且 } |E_1| = p^k \in \mathbb{Z}_{\geq 2}$$

$$\text{任取 } d \in E_1 \quad \therefore d \in G \text{ 且 } Stab_G(E_1) \leq G \quad \therefore Stab_G(E_1) \cdot d \subseteq G$$

$$\text{任取 } Stab_G(E_1) \cdot d \text{ 中的一元: } \varepsilon d \text{ (其中 } \varepsilon \in Stab_G(E_1) \text{)}$$

$$\therefore \varepsilon \in Stab_G(E_1) \quad \therefore \varepsilon \in G \text{ 且 } \varepsilon E_1 = E_1 \quad \equiv \therefore \varepsilon d \in \varepsilon E_1 = E_1$$

$$\therefore Stab_G(E_1) \cdot d \subseteq E_1 \quad \text{~~G是群~~}$$

$$\therefore G \text{ 是有限群, } d \in G, Stab_G(E_1) \leq G \quad \equiv$$

$$\therefore Stab_G(E_1) \cdot d \subseteq G \text{ 且 } |Stab_G(E_1) \cdot d| = |Stab_G(E_1)|$$

$$\therefore p^k \leq |Stab_G(E_1)| = |Stab_G(E_1) \cdot d| \leq |E_1| = p^k$$

$$\therefore |Stab_G(E_1)| = p^k \text{ 且 } Stab_G(E_1) \leq G$$

$$\therefore G \text{ 中有阶为 } p^k \text{ 的子群 } Stab_G(E_1) \quad \square$$

定义(群的关于包含关系极大的  $p$ -子群)  $G$  是群,  $p$  是素数,  $P \leq G$ ,  $P$  是  $G$  的  $p$ -子群. 如果对  $G$  的任意  $p$ -子群  $Q \leq G$  都有  $(P \leq Q \Rightarrow Q = P)$ , 则称  $G$  的  $p$ -子群  $P$  是  $G$  的一个关于包含关系极大的  $p$ -子群.

Lemma:  $G$  是群,  $\text{Sub}(G) = \{K \leq G \mid K \leq G\}$ . 则有:

① 映射  $\alpha: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$  是群  $G$  在非空集合  $\text{Sub}(G)$  上  $(g, K) \mapsto gKg^{-1}$

的群作用

② 对  $\forall K \leq G$ , 有:  $K$  在上述群作用下的  $G$ -轨道为

$$X_K := GK = \{gKg^{-1} \mid g \in G\}. \text{ 且有 } \text{Stab}_G(K) = N_G(K)$$

③ 映射  $\varphi: G/N_G(K) \rightarrow X_K$  是  $G$ -集间的同构.  
 $gN_G(K) \mapsto gKg^{-1}$

因此有  $|X_K| = (G : N_G(K)) \rightarrow$  参见轨道-稳定化子定理.

④ 对  $\forall H \leq G$ , 有:  $X_K$  在  $H$  的共轭作用下保持不变. 且  
 对  $\forall Q \in X_K$ , 都有  $Q \in X_K^H \Leftrightarrow H \leq N_G(Q)$

Proof:  $\because G \leq G, G \leq G \therefore G \in \text{Sub}(G) \therefore \text{Sub}(G) \neq \emptyset$   
 $\therefore \text{Sub}(G)$  是非空集合.

对  $\forall (g, K) \in G \times \text{Sub}(G)$ , 有:  $g \in G$  且  $K \leq G$

$$\therefore a(g, K) = gKg^{-1} \leq G \quad \therefore a(g, K) \in \text{Sub}(G)$$

对  $\forall (g_1, K_1), (g_2, K_2) \in G \times \text{Sub}(G)$ , 若  $(g_1, K_1) = (g_2, K_2)$

则  $g_1 = g_2 \in G$  且  $K_1 = K_2 \leq G$

$$\therefore a(g_1, K_1) = g_1 K_1 g_1^{-1} = g_2 K_2 g_2^{-1} = a(g_2, K_2)$$

↓  
双方向包含显然.

$\therefore a: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$  是良定义的映射.

$$\text{对 } \forall K \leq G, \text{ 有: } a(1_G, K) = 1_G K 1_G^{-1} = 1_G K 1_G = K$$

对  $\forall g_1, g_2 \in G, \forall K \leq G$ , 有:  $g_1 g_2 \in G$ .

$$a(g_1, a(g_2, K)) = a(g_1, g_2 K g_2^{-1}) = g_1 (g_2 K g_2^{-1}) g_1^{-1}$$

$$= (g_1 g_2) K (g_2^{-1} g_1^{-1}) = (g_1 g_2) K (g_1 g_2)^{-1} = a(g_1 g_2, K)$$

↓  
双方向包含显然

$\therefore$  映射  $a: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$  是群  $G$  在非空集合  $\text{Sub}(G)$  上的

$$(g, K) \mapsto gKg^{-1}$$

群作用. ①得证.

对  $\forall K \in \text{Sub}(G)$ , 有:  $K \leq G$ .  $\therefore K$  所在的  $G$ -轨道  $GK$  为:

$$GK = \{gKg^{-1} \mid g \in G\} \subseteq \text{Sub}(G).$$

$$\therefore X_K = GK = \{gKg^{-1} \mid g \in G\} \subseteq \text{Sub}(G)$$

$K$  在  $G$  中的稳定化子为  $\text{Stab}_G(K) = \{g \in G \mid gKg^{-1} = K\} = N_G(K)$

$\therefore \text{Stab}_G(K) = N_G(K) \leq G$  . ②得证.

又  $\forall K \leq G$ , 有:  $\because K \leq G \quad \therefore N_G(K) \leq G$ ,  $G/N_G(K) = \{gN_G(K) \mid g \in G\}$

定义映射  $\alpha: G \times G/N_G(K) \longrightarrow G/N_G(K)$   
 $= \{xN_G(K) \mid x \in G\}$

$$(g, xN_G(K)) \longmapsto (gx)N_G(K)$$

则已经证明了映射  $\alpha$  是群  $G$  在  $G/N_G(K)$  上的群作用.

$\because G$  是群,  $K \leq G$ ,  $X_K = \{gKg^{-1} \mid g \in G\} = \{xKx^{-1} \mid x \in G\}$ ,  $G \leq G$ ,

定义映射  $\alpha: G \times X_K \longrightarrow X_K$

$$(g, xKx^{-1}) \longmapsto (gx)K(x^{-1}g^{-1})$$

则已经证明了映射  $\alpha$  是群  $G$  在非空集合  $X_K$  上的群作用.

定义映射  $\varphi: G/N_G(K) \longrightarrow X_K$

$$xN_G(K) \longmapsto xKx^{-1}$$

则有: 对  $\forall xN_G(K) \in G/N_G(K)$  (其中  $x \in G$ )  $\because x \in G \quad \therefore xKx^{-1} \in X_K$

$\therefore \varphi(xN_G(K)) = xKx^{-1} \in X_K \quad \therefore \varphi(G/N_G(K)) \subseteq X_K$ .

对  $\forall xN_G(K), yN_G(K) \in G/N_G(K)$  (其中  $x, y \in G$ ).

若  $xN_G(K) = yN_G(K)$ , 则有:  $\because K \leq G \quad \therefore N_G(K) \leq G \quad \therefore y^{-1}x \in N_G(K)$

$\therefore y^{-1}x \in G$  且  $(y^{-1}x)K(y^{-1}x)^{-1} = K \quad \therefore (y^{-1}x)K(x^{-1}y) = K$

$$\therefore xKx^{-1} = yKy^{-1}$$

$$(\nexists \forall x k x^{-1} \in x K x^{-1} \text{ (其中 } k \in K) \therefore k \in K \leq G$$

$$\therefore (y^{-1}x)k(x^{-1}y) \in (y^{-1}x)K(x^{-1}y) = K \therefore \exists \beta \in K, \text{ s.t.}$$

$$(y^{-1}x)k(x^{-1}y) = \beta \therefore y^{-1}(x k x^{-1})y = \beta$$

$$\therefore x k x^{-1} = y \beta y^{-1} \in y K y^{-1} \therefore x K x^{-1} \subseteq y K y^{-1}$$

$$\nexists \forall y k y^{-1} \in y K y^{-1} \text{ (其中 } k \in K) \therefore k \in K = (y^{-1}x)K(x^{-1}y)$$

$$\therefore \exists \beta \in K, \text{ s.t. } k = (y^{-1}x)\beta(x^{-1}y) = y^{-1}(x\beta x^{-1})y$$

$$\therefore y k y^{-1} = x \beta x^{-1} \in x K x^{-1} \therefore y K y^{-1} \subseteq x K x^{-1}$$

$$\therefore x K x^{-1} = y K y^{-1} )$$

$$\therefore \varphi(x N_G(K)) = x K x^{-1} = y K y^{-1} = \varphi(y N_G(K))$$

$\therefore \varphi$  是良定义的映射

$$\text{若 } \varphi(x N_G(K)) = \varphi(y N_G(K)), \text{ 则 } x K x^{-1} = y K y^{-1}$$

$$\therefore x, y \in G \therefore x^{-1}y \in G. \therefore (x^{-1}y)K(y^{-1}x) = K$$

$$(\nexists \forall (x^{-1}y)k(y^{-1}x) \in (x^{-1}y)K(y^{-1}x), \text{ 其中 } k \in K.$$

$$\therefore k \in K \therefore \cancel{x k x^{-1}} y k y^{-1} \in y K y^{-1} = x K x^{-1} \therefore \exists \beta \in K, \text{ s.t.}$$

$$y k y^{-1} = x \beta x^{-1} \therefore (x^{-1}y)k(y^{-1}x) = x^{-1}(y k y^{-1})x = x^{-1}(x \beta x^{-1})x$$

$$= (x^{-1}x)\beta(x^{-1}x) = 1_G \beta 1_G = \beta \in K \therefore (x^{-1}y)K(y^{-1}x) \subseteq K$$

$$\nexists \forall k \in K \therefore k \in K \therefore x k x^{-1} \in x K x^{-1} = y K y^{-1} \therefore \exists \beta \in K, \text{ s.t.}$$

$$x k x^{-1} = y \beta y^{-1} \therefore k = x^{-1}(y \beta y^{-1})x = (x^{-1}y)\beta(y^{-1}x) \in (x^{-1}y)K(y^{-1}x)$$

$$\therefore K \subseteq (x^{-1}y)K(y^{-1}x) \quad \therefore (x^{-1}y)K(y^{-1}x) = K$$

$$\therefore x^{-1}y \in G \text{ 且 } (x^{-1}y)K(x^{-1}y)^{-1} = K \quad \therefore x^{-1}y \in N_G(K)$$

$$\therefore xN_G(K) = yN_G(K) \quad \therefore \varphi \text{ 是单射.}$$

$$\text{对 } \forall xKx^{-1} \in X_K \text{ (其中 } x \in G \text{)} \quad \therefore x \in G \quad \therefore xN_G(K) \in G/N_G(K)$$

$$\therefore \varphi(xN_G(K)) = xKx^{-1} \quad \therefore \varphi \text{ 是满射.} \quad \therefore \varphi \text{ 是双射.}$$

~~$$\text{对 } \forall g \in G, \forall xKx^{-1} \in X_K \text{ (其中 } x \in G \text{), 有: } gx \in G$$~~

~~$$\varphi(a(g, xKx^{-1})) = \varphi((gx)K(x^{-1}g^{-1})) = \varphi((gx)K(gx)^{-1})$$~~

$$\text{对 } \forall g \in G, \forall xN_G(K) \in G/N_G(K) \text{ (其中 } x \in G \text{), 有: } gx \in G$$

$$\therefore \varphi(a(g, xN_G(K))) = \varphi((gx)N_G(K)) = (gx)K(gx)^{-1}$$

$$= (gx)K(x^{-1}g^{-1}) = a(g, xKx^{-1}) = a(g, \varphi(xN_G(K)))$$

$$\therefore \varphi: G/N_G(K) \rightarrow X_K \text{ 是 } G\text{-集之间的同构.}$$

$$\therefore |X_K| = |G/N_G(K)| = (G:N_G(K)) \quad \text{③得证.}$$

$$\text{对 } \forall K \leq G, \forall H \leq G, \text{ 有: } X_K = \{gKg^{-1} | g \in G\} = \{xKx^{-1} | x \in G\}$$

$$\text{定义映射 } \alpha: H \times X_K \longrightarrow X_K$$

$$(g, xKx^{-1}) \longmapsto (gx)K(x^{-1}g^{-1})$$

则已经证明了 映射  $\alpha$  是群  $H$  在非空集合  $X_K$  上的群作用.

对  $\forall h \in H, \forall xKx^{-1} \in X_K$  (其中  $x \in G$ ) 有:

$$\because x \in G, K \leq G \quad \therefore xKx^{-1} \leq G \quad \therefore xKx^{-1} \in \text{Sub}(G)$$

按照①中的群作用  $\alpha$ , 有:  $h \in H \leq G, xKx^{-1} \in \text{Sub}(G)$

$$\therefore \alpha(h, xKx^{-1}) = h(xKx^{-1})h^{-1} = (hx)K(x^{-1}h^{-1})$$

$\downarrow$   
双方向包含显然

$$= (hx)K(hx)^{-1} \in X_K \quad \therefore X_K \text{ 在 } H \text{ 的共轭作用下保持不变.}$$

对  $\forall Q \in X_K$ , 有:  $Q \in X_K^H \Leftrightarrow$  对  $\forall h \in H$ , 都有  $hQh^{-1} = Q$

$$\Leftrightarrow \text{对 } \forall h \in H, \text{ 都有 } h \in N_G(Q) \Leftrightarrow H \subseteq N_G(Q) \Leftrightarrow H \leq N_G(Q)$$

④得证.  $\square$

$$\text{Remark: } \because Q \in X_K \subseteq \text{Sub}(G) \quad \therefore Q \leq G \quad \therefore N_G(Q) \leq G$$

Lemma:  $G$  是群,  $K \leq G$ ,  $p$  是素数,  $(G:K) \in \mathbb{Z}_{\geq 1}, p \nmid (G:K)$ .

令  $X_K = \{gKg^{-1} \mid g \in G\}$  则有:  $X_K$  是非空有限集, 并且

$$p \nmid |X_K|.$$

$$\text{Proof: } \because G \text{ 是群}, K \leq G \quad \therefore K \triangleleft N_G(K) \leq G \quad \therefore K \leq N_G(K) \leq G$$

$$\therefore (G:K) = \cancel{G:N_G(K)} (G:N_G(K)) \cdot (N_G(K):K)$$

$$\because G \text{ 是群}, K \leq G, X_K = \{gKg^{-1} \mid g \in G\} \quad \therefore |X_K| = (G:N_G(K))$$

$$\therefore (G:K) = |X_K| \cdot (N_G(K):K) \quad \therefore (G:K) \in \mathbb{Z}_{\geq 1}$$

$\therefore |X_K| \in \mathbb{Z}_{\geq 1}$  且  $(N_G(K):K) \in \mathbb{Z}_{\geq 1} \quad \therefore X_K$  是非空有限集合.

假设  $p \mid |X_K|$ , 则  $p \mid (G:K)$ . 与  $p \nmid (G:K)$  矛盾.  $\therefore p \nmid |X_K|$

$\therefore |X_K| \in \mathbb{Z}_{\geq 1}$ ,  $X_K$  是非空有限集合,  $p \nmid |X_K|$   $\square$

Lemma:  $H$  是有限  $p$ -群 ( $p$  是素数),  $X$  是非空有限集合, 群  $H$  作用在集合  $X$  上. 若  $p \nmid |X|$ , 则有  $X^H \neq \emptyset$ .

Proof:  $\because X$  是非空有限集合  $\therefore |X| \in \mathbb{Z}_{\geq 1}$  假设  $X^H = \emptyset$ , 则

$|X^H| = 0 \quad \because H$  是有限  $p$ -群,  $X$  是非空有限集合

$\therefore |X| \equiv |X^H| \pmod{p} \quad \therefore |X| \equiv 0 \pmod{p} \quad \therefore p \mid |X|$

矛盾.  $\therefore X^H \neq \emptyset$ .  $\square$

Lemma:  $G$  是群,  $P, H \leq G$  都是  $G$  的  $p$ -子群 ( $p$  是素数),  $P$  是  $G$  的一个关于包含关系极大的  $p$ -子群, ~~重~~  $HP \leq G$ , 则有:

$H \leq P$

Proof:  $\because P, H \leq G$  都是  $G$  的  $p$ -子群  $\therefore \exists a, b \in \mathbb{Z}_{\geq 0}$ , s.t.

$|H| = p^a$  且  $|P| = p^b \quad \because H, P \leq G \quad \therefore H \cap P \leq G$

$\because H \cap P \leq G, H \leq G, H \cap P \subseteq H \quad \therefore H \cap P \leq H$

$\because H$  是  $p$ -群,  $H \cap P \leq H$   $\because H \cap P$  是  $p$ -群

$\because \exists c \in \mathbb{Z}_{\geq 0}$ , s.t.  $|H \cap P| = p^c$   $\because H$  是有限群,  $H \cap P \leq H$

$\because |H \cap P| \mid |H|$   $\because p^c \mid p^a$   $\because c \leq a$   $\because a - c \geq 0$

$\because G$  是群,  $H \leq G$ ,  $P \leq G$   $\because |HP| \cdot |H \cap P| = |H| \cdot |P|$

$\because |HP| \cdot p^c = p^a \cdot p^b = p^{a+b}$   $\because a+b \in \mathbb{Z}_{\geq 0}$   $\because p^{a+b} \in \mathbb{Z}_{\geq 1}$

$\because c \in \mathbb{Z}_{\geq 0}$   $\because p^c \in \mathbb{Z}_{\geq 1}$   $\because |HP| \in \mathbb{Z}_{\geq 1}$   $\because |HP| = p^{a+b-c}$

$\because a+b-c = b+(a-c) \geq 0$   $\because |HP| = p^{a+b-c} \in \mathbb{Z}_{\geq 1}$ ,  $a+b-c \in \mathbb{Z}_{\geq 0}$

$\because HP \leq G$   $\because HP$  是  $G$  的  $p$ -子群  $\because P \subseteq HP$   $\because P \leq HP$

$\because P$  是  $G$  的一个关于包含关系极大的  $p$ -子群  $\because HP = P$

$\because H \subseteq HP = P$   $\because H \subseteq P$   $\because H \leq P$   $\square$

Lemma:  $G$  是群,  $P$  是  $G$  的一个关于包含关系极大的  $p$ -子群,  $H \leq G$  是  $G$  的  $p$ -子群,  $H \leq N_G(P)$ , 则有  $H \leq P$

Proof:  $\because P$  是  $G$  的一个关于包含关系极大的  $p$ -子群

$\because P$  是  $G$  的  $p$ -子群,  $\because P \leq G$   $\because N_G(P) \leq G$ .

$\because G$  是群,  $H \leq G$ ,  $P \leq G$ ,  $H \leq N_G(P)$   $\because HP = PH$  且  $HP \leq G$

$\because H \leq P$ .  $\square$

Lemma:  $G$  是有限群,  $p$  是素数,  $P$  是  $G$  的 Sylow  $p$ -子群, 则  $P$  是  $G$  的一个关于包含关系极大的  $p$ -子群.

Proof:  $\because G$  是有限群  $\therefore |G| \in \mathbb{Z}_{\geq 1}$   $\therefore$  设  $|G| = p^m \cdot a$ , 其中

$m \in \mathbb{Z}_{\geq 0}$ ,  $a \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid a$   $\therefore p^m \parallel |G|$

$\because P$  是  $G$  的 Sylow  $p$ -子群  $\therefore P \leq G$  且  $|P| = p^m$  (其中  $m \in \mathbb{Z}_{\geq 0}$ )

$\therefore P$  是  $G$  的  $p$ -子群. 对  $G$  的任意  $p$ -子群  $Q \leq G$ , 如果  $P \leq Q$ ,

则有:  $\because Q \leq G$  且  $Q$  是  $G$  的  $p$ -子群  $\therefore |Q| = p^r$  (其中  $r \in \mathbb{Z}_{\geq 0}$ )

$\because G$  是有限群,  $Q \leq G$   $\therefore |Q| \mid |G|$   $\therefore p^r \mid p^m \cdot a$

$\because p \nmid a$   $\therefore p^{m+1} \nmid |G|$   $\therefore p^{m+1} \nmid p^m \cdot a$   $\therefore r \leq m$

(若假设  $r > m$ , 则  $r \in \mathbb{Z}_{\geq 0}$  且  $m \in \mathbb{Z}_{\geq 0}$  且  $r > m$   $\therefore r \geq m+1$

$\therefore p^{m+1} \mid p^r$ ,  $p^r \mid p^m \cdot a$   $\therefore p^{m+1} \mid p^m \cdot a$  矛盾.  $\therefore r \leq m$ )

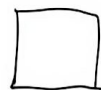
$\therefore |Q| = p^r$ ,  $r \in \mathbb{Z}_{\geq 0}$   $\therefore |Q| = p^r \in \mathbb{Z}_{\geq 1}$   $\therefore Q$  是有限群.

$\because Q$  是有限群,  $P \leq Q$   $\therefore |P| \mid |Q|$   $\therefore p^m \mid p^r$   $\therefore m \leq r$

$\therefore r \leq m \leq r$   $\therefore m = r$   $\therefore |P| = p^m = p^r = |Q|$

$\because Q$  是有限群,  $P \leq Q$ ,  $|P| = |Q|$   $\therefore P = Q$   $\therefore Q = P$

$\therefore P$  是  $G$  的一个关于包含关系极大的  $p$ -子群



Lemma:  $G$  是有限群,  $P$  是  $G$  的 Sylow  $p$ -子群,  $H$  是  $G$  的  $p$ -子群,

$H \leq N_G(P)$ , 则有:  $H \leq P$

Proof:  $\because G$  是有限群,  $P$  是  $G$  的 Sylow  $p$ -子群

$\therefore P$  是  $G$  的一个关于包含关系极大的  $p$ -子群  $\therefore H \leq P \quad \square$

Lemma:  $G$  是群,  $P$  是  $G$  的一个关于包含关系极大的  $p$ -子群 ( $p$  是素数),

记  $X_P = \{gPg^{-1} \mid g \in G\}$ . 定义映射  $\alpha: P \times X_P \longrightarrow X_P$   
 $(g, xPx^{-1}) \mapsto (gx)P(x^{-1}g^{-1})$

则已经证明了映射  $\alpha$  是群  $P$  在非空集合  $X_P$  上的群作用.

则有:  $X_P^P = \{P\}$ .

Proof:  $\because I_G \in G$  且  $I_G P I_G^{-1} = I_G P I_G = \{I_G x I_G \mid x \in P\} = \{x \mid x \in P\}$

$= P$ . 且已知  $P \leq G \quad \therefore P \in X_P$

对  $\forall g \in P$ , 都有  $\alpha(g, P) = \alpha(g, I_G P I_G^{-1}) = (g I_G) P (I_G^{-1} g^{-1})$

$= g P g^{-1} = P$

( $\because G$  是群,  $P \leq G \quad \therefore P \triangleleft N_G(P) \quad \therefore P \subseteq N_G(P)$ )

$\therefore$  对  $\forall g \in P$ , 有:  $g \in N_G(P) \quad \therefore g \in G$  且  $g P g^{-1} = P$  )

$\therefore P \in X_P^P \quad \therefore \{P\} \subseteq X_P^P$

对  $\forall Q \in X_P^P$ , 有:  $\because Q \in X_P^P \quad \therefore Q \in X_P \quad \therefore \exists x \in G, \text{ s.t. }$

$$Q = xPx^{-1}. \because G \text{ 是群}, P \leq G, x \in G \therefore xPx^{-1} \leq G$$

$$\therefore Q \leq G \quad \therefore |Q| = |xPx^{-1}| = |P| \quad \therefore P \text{ 是 } G \text{ 的 } p\text{-子群}$$

$$\therefore Q \text{ 是 } G \text{ 的 } p\text{-子群}. \quad \because P \text{ 是有限群}, Q \text{ 也是有限群}.$$

$$\text{对 } \forall g \in P, \therefore Q \in X_P^P \quad \therefore a(g, Q) = Q \quad \therefore gQg^{-1} = Q$$

$$\therefore g \in G \text{ 且 } gQg^{-1} = Q, Q \leq G \quad \therefore g \in N_G(Q) \quad \therefore P \subseteq N_G(Q)$$

$$\therefore N_G(Q) \leq G \quad \therefore P \leq N_G(Q)$$

已知  $G$  是群,  $p$  是素数,  $Q \leq G$ ,  $Q$  是  $G$  的  $p$ -子群. 对  $G$  的任意  $p$ -子群  $R \leq G$ , 若  $Q \leq R$ , 则  $\therefore Q \leq R \therefore xPx^{-1} \leq R$ .

$$\therefore x \in G \quad \therefore x^{-1} \in G \quad \therefore R \leq G \text{ 且 } x^{-1} \in G \quad \therefore x^{-1}R(x^{-1})^{-1} \leq G$$

$$\therefore x^{-1}Rx \leq G \text{ 且 } P \leq G. \quad \text{对 } \forall t \in P, \therefore xt x^{-1} \in xPx^{-1} \subseteq R$$

$$\therefore \exists r \in R, \text{ s.t. } xt x^{-1} = r \quad \therefore x \in G, t \in G, r \in G$$

$$\therefore t = x^{-1}rx \in x^{-1}Rx \quad \therefore P \subseteq x^{-1}Rx \quad \therefore P \leq x^{-1}Rx$$

$$\therefore G \text{ 是群}, R \leq G, x^{-1} \in G \quad \therefore |x^{-1}Rx| = |R| \quad \therefore R \text{ 是 } p\text{-群}$$

$$\therefore x^{-1}Rx \text{ 是 } p\text{-群} \quad \therefore x^{-1}Rx \leq G \text{ 是 } G \text{ 的 } p\text{-子群且 } P \leq x^{-1}Rx$$

$$\therefore P \text{ 是 } G \text{ 的一个关于包含关系极大的 } p\text{-子群} \quad \therefore x^{-1}Rx = P$$

$$\therefore Q = xPx^{-1} = x(x^{-1}Rx)x^{-1} = (xx^{-1})R(xx^{-1}) = R \quad \therefore R = Q$$

↓  
双方向包含显然.

$\therefore G$  的  $p$ -子群  $Q$  是  $G$  的一个关于包含关系极大的  $p$ -子群.

$\because G$  是群,  $Q$  是  $G$  的一个关于包含关系极大的  $p$ -子群,  $P \leq G$  是  $G$  的  $p$ -子群,  $P \leq N_G(Q)$ .  $\therefore P \leq Q$ .  $\therefore P \subseteq Q$

$\because Q$  是有限群,  $P \leq Q$ ,  $|P| = |Q| \therefore Q = P \therefore Q \in \{P\}$

$\therefore X_P^P \subseteq \{P\} \therefore X_P^P = \{P\} \quad \square$

定理 (Sylow 第二、第三联合定理)  $G$  是有限群,  $p$  是素数, 设  $|G| = p^m \cdot k$ , 其中  $m \in \mathbb{Z}_{\geq 0}$ ,  $k \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid k$ . 记  $G$  的所有 Sylow  $p$ -子群组成的集合为  $\text{Syl}_p(G)$ . 记  $n_p = |\text{Syl}_p(G)|$ , 则有:

①  $G$  的任意一个  $p$ -子群  $H \leq G$  都包含于  $G$  的某个 Sylow  $p$ -子群

② 对  $\forall P \in \text{Syl}_p(G)$ , 都有:  $\text{Syl}_p(G) = \{xPx^{-1} \mid x \in G\}$

③ 对  $\forall P_1, P_2 \in \text{Syl}_p(G)$ ,  $\exists g \in G$ , s.t.  $P_2 = gP_1g^{-1}$

④ 对  $\forall P \in \text{Syl}_p(G)$ , 都有:  $n_p = (G : N_G(P))$

⑤  $n_p \mid k$

⑥ 对  $\forall P \in \text{Syl}_p(G)$ , 都有  $n_p \mid (G : P)$

⑦  $n_p \equiv 1 \pmod{p}$

Proof:  $\because G$  是有限群  $\therefore |G| \in \mathbb{Z}_{\geq 1} \therefore p$  是素数,  $|G| \in \mathbb{Z}_{\geq 1}$

$\therefore$  存在唯一的  $v_p(|G|) \in \mathbb{Z}_{\geq 0}$ , s.t.  $p^{v_p(|G|)} \parallel |G|$ . 记  $m = v_p(|G|) \in \mathbb{Z}_{\geq 0}$

$\therefore m \in \mathbb{Z}_{\geq 0}$  且  $p^m \parallel |G|$   $\therefore p^m \mid |G|$  且  $p^{m+1} \nmid |G|$   $\therefore m \in \mathbb{Z}_{\geq 0} \therefore p^m \in \mathbb{Z}_{\geq 1}$

$\therefore p^m \mid |G| \therefore \frac{|G|}{p^m} \in \mathbb{Z}_{\geq 1}$  记  $k = \frac{|G|}{p^m} \therefore |G| = p^m \cdot k, k \in \mathbb{Z}_{\geq 1}$

假设  $p \mid k$ , 则  $\frac{k}{p} \in \mathbb{Z}_{\geq 1} \therefore \frac{p^m \cdot k}{p^{m+1}} \in \mathbb{Z}_{\geq 1} \therefore p^{m+1} \mid p^m \cdot k$

$\therefore p^{m+1} \mid |G|$  矛盾.  $\therefore p \nmid k$

$\therefore |G| \in \mathbb{Z}_{\geq 1}, |G| = p^m \cdot k, m \in \mathbb{Z}_{\geq 0}, k \in \mathbb{Z}_{\geq 1}, p \nmid k$

$\therefore G$  是有限群,  $p$  是素数,  $m \in \mathbb{Z}_{\geq 0}, p^m \mid |G| \therefore G$  中一定有为  $p^m$  的子群.  $\therefore p^m \parallel |G| \therefore G$  中的阶为  $p^m$  的子群是  $G$  的 Sylow  $p$ -子群.

$\therefore \text{Syl}_p(G) \neq \emptyset$ .

设  $H$  是  $G$  的任意一个  $p$ -子群.  $\therefore H \leq G$  且  $H$  是  $p$ -群.

对  $\forall P \in \text{Syl}_p(G)$ , 记  $X = \{xPx^{-1} \mid x \in G\}$

任取  $X$  中的一个元素  $xPx^{-1}$  (其中  $x \in G$ ).  $\therefore P \in \text{Syl}_p(G) \therefore P \leq G$

$\therefore G$  是群,  $P \leq G, x \in G \therefore xPx^{-1} \leq G$  且  $|xPx^{-1}| = |P| = p^m$

$\therefore xPx^{-1}$  是  $G$  的 Sylow  $p$ -子群  $\therefore xPx^{-1} \in \text{Syl}_p(G) \therefore X \subseteq \text{Syl}_p(G)$

$\therefore G$  是有限群,  $P \leq G \therefore |G| = |P| \cdot (G:P)$

$\therefore (G:P) = \frac{|G|}{|P|} = \frac{p^m \cdot k}{p^m} = k \therefore p \nmid k \therefore p \nmid (G:P)$

$\therefore G$  是有限群,  $P \leq G, p$  是素数,  $(G:P) = k \in \mathbb{Z}_{\geq 1}, p \nmid (G:P)$

$X = \{xPx^{-1} \mid x \in G\} \therefore X$  是非空有限集, 并且  $p \nmid |X|$

$\therefore H$  是有限群,  $H$  是  $p$ -群,  $X$  是非空有限集合

$\therefore$  一定至少存在一个群  $H$  在非空有限集合  $X$  上的群作用.

$\therefore G$  是有限群,  $P \leq G$ ,  $X = \{xPx^{-1} \mid x \in G\}$ ,  $H \leq G$

$\therefore$  定义群作用  $\alpha: H \times X \longrightarrow X$  则  $\alpha$  是群  $H$  在  
 $(g, xPx^{-1}) \mapsto (gx)P(x^{-1}g^{-1})$

非空有限集合  $X$  上的群作用.

$\therefore p \nmid |X| \therefore X^H \neq \emptyset \therefore \exists Q \in X^H \therefore Q \in X$

对  $\forall h \in H$ ,  $\therefore Q \in X^H \therefore \alpha(h, Q) = Q \therefore hQh^{-1} = Q$

$\therefore h \in G$  且  $hQh^{-1} = Q$  且  $Q \leq G \therefore h \in N_G(Q) \therefore H \leq N_G(Q)$

$\therefore H \leq G, N_G(Q) \leq G \therefore H \leq N_G(Q) \therefore Q \in X \subseteq \text{Syl}_p(G)$

$\therefore Q$  是  $G$  的 Sylow  $p$ -子群.

$\therefore G$  是有限群,  $Q$  是  $G$  的 Sylow  $p$ -子群,  $H$  是  $G$  的  $p$ -子群,  $H \leq N_G(Q)$

$\therefore H \leq Q$ . ①得证.  $\therefore Q \in X \therefore \exists \alpha \in G, \text{s.t. } Q = \alpha P \alpha^{-1} \therefore H \leq \alpha P \alpha^{-1}$

对  $\forall P \in \text{Syl}_p(G)$ , 记  $X = \{xPx^{-1} \mid x \in G\}$  则已经证明

$X \subseteq \text{Syl}_p(G)$  对  $\forall R \in \text{Syl}_p(G)$ , 有:  $R$  是  $G$  的 Sylow  $p$ -子群

$\therefore R$  是  $G$  的  $p$ -子群.

$\because G$  是有限群,  $P \leq G$ ,  $X = \{xPx^{-1} \mid x \in G\}$ ,  $R \leq G$

$\therefore$  定义群作用  $\alpha: R \times X \longrightarrow X$  则映射  $\alpha$  是群  $R$  在  
 $(g, xPx^{-1}) \mapsto (gx)P(x^{-1}g^{-1})$

非空集合  $X$  上的群作用.

$\because G$  是有限群,  $P \leq G$   $\therefore |G| = |P|(G:P)$

$\therefore (G:P) = \frac{|G|}{|P|} = \frac{p^m \cdot k}{p^m} = k$   $\because p \nmid k$   $\therefore p \nmid (G:P)$

$\because G$  是群,  $P \leq G$ ,  $p$  是素数,  $(G:P) = k \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid (G:P)$

$X = \{xPx^{-1} \mid x \in G\}$   $\therefore X$  是非空有限集合, 并且  $p \nmid |X|$

$\because R$  是有限群,  $R$  是  $p$ -群,  $X$  是非空有限集合,  $p \nmid |X|$   $\therefore X^R \neq \emptyset$

$\therefore \exists D \in X^R$   $\therefore D \in X$   $\therefore D \in X \subseteq \text{Syl}_p(G)$   $\therefore D \leq G$

对  $\forall r \in R$ ,  $\because D \in X^R$   $\therefore \alpha(r, D) = D$   $\therefore rDr^{-1} = D$

$\therefore r \in G$  且  $rDr^{-1} = D$  且  $D \leq G$   $\therefore r \in N_G(D)$   $\therefore R \subseteq N_G(D)$

$\therefore R \leq N_G(D)$   $\therefore R \leq D$   $\because D \in X$

$\because D$  是有限群,  $R \leq D$ ,  $|R| = |D| = p^m$   $\therefore R = D \in X$

$\therefore \text{Syl}_p(G) \subseteq X$   $\therefore \text{Syl}_p(G) = X = \{xPx^{-1} \mid x \in G\}$  ②得证.

对  $\forall P_1, P_2 \in \text{Syl}_p(G)$ , 有: 任取  $P \in \text{Syl}_p(G)$ , 有:  ~~$\text{Syl}_p(G)$~~

$\text{Syl}_p(G) = \{xPx^{-1} \mid x \in G\}$   $\therefore \exists x_1, x_2 \in G$ , s.t.  $P_1 = x_1Px_1^{-1}$

$P_2 = x_2Px_2^{-1}$   $\therefore x_2x_1^{-1} \in G$  且  $(x_2x_1^{-1})P_1(x_2x_1^{-1})^{-1} = x_2x_1^{-1}P_1x_1x_2^{-1}$

$$= x_2 x_1^{-1} x_1 P x_1^{-1} x_1 x_2^{-1} = x_2 P x_2^{-1} = P_2 \quad \text{③得证}$$

$$\text{对 } \forall P \in \text{Syl}_p(G), \text{ 都有: } \text{Syl}_p(G) = \{xPx^{-1} \mid x \in G\}$$

$$\because P \leq G \quad \therefore |\text{Syl}_p(G)| = |\{xPx^{-1} \mid x \in G\}| = (G : N_G(P))$$

$$\therefore n_p = |\text{Syl}_p(G)| = (G : N_G(P)) \quad \text{④得证}$$

$$\text{对 } \forall P \in \text{Syl}_p(G), \text{ 有: } |P| = p^m, (G : P) = \frac{|G|}{|P|} = k \in \mathbb{Z}_{\geq 1}$$

$$P \triangleleft N_G(P) \leq G \quad \therefore (G : P) = (G : N_G(P)) (N_G(P) : P)$$

$$\therefore k = n_p \cdot (N_G(P) : P) \quad \because k \in \mathbb{Z}_{\geq 1}, n_p \in \mathbb{Z}_{\geq 1} \quad \therefore (N_G(P) : P) \in \mathbb{Z}_{\geq 1}$$

$$\therefore n_p \mid k \quad \text{⑤得证} \quad \therefore n_p \mid (G : P) \quad \text{⑥得证}$$

$$\text{对 } \forall P \in \text{Syl}_p(G), \text{ 有: } \text{Syl}_p(G) = \{xPx^{-1} \mid x \in G\}.$$

$$\because P \text{ 是 } G \text{ 的 Sylow } p\text{-子群} \quad \therefore P \text{ 是 } G \text{ 的一个关于包含关系极大的 } p\text{-子群}$$

$$\text{定义映射 } \alpha: P \times \text{Syl}_p(G) \longrightarrow \text{Syl}_p(G) \quad \text{则已经证明}$$

$$(g, xPx^{-1}) \longmapsto (gx)P(x^{-1}g^{-1})$$

$$\text{映射 } \alpha \text{ 是群 } P \text{ 在非空集合 } \text{Syl}_p(G) \text{ 上的群作用}$$

$$\text{则有: } \text{Syl}_p(G)^P = \{P\} \quad \therefore |\text{Syl}_p(G)^P| = 1$$

$$\therefore |\text{Syl}_p(G)| \equiv |\text{Syl}_p(G)^P| \pmod{p}$$

$$\therefore n_p \equiv 1 \pmod{p} \quad \text{⑦得证} \quad \square$$

接下来对 Sylow 第二、第三定理的证明进行第三代重构。

Lemma:  $G$  是有限群,  $p$  是素数, 则有:  $|G| = p^m \cdot k$ , 其中  $m \in \mathbb{Z}_{\geq 0}$ ,  $k \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid k$ .

Proof:  $\because G$  是有限群  $\therefore |G| \in \mathbb{Z}_{\geq 1}$   $\because p$  是素数,  $|G| \in \mathbb{Z}_{\geq 1}$

$\therefore$  存在唯一的  $v_p(|G|) \in \mathbb{Z}_{\geq 0}$ , s.t.  $p^{v_p(|G|)} \parallel |G|$ .

记  $m = v_p(|G|) \in \mathbb{Z}_{\geq 0}$   $\therefore m \in \mathbb{Z}_{\geq 0}$  且  $p^m \parallel |G|$

$\therefore p^m \parallel |G|$  且  $p^{m+1} \nmid |G|$   $\because m \in \mathbb{Z}_{\geq 0}$   $\therefore p^m \in \mathbb{Z}_{\geq 1}$

$\therefore p^m \parallel |G|$ ,  $p^m \in \mathbb{Z}_{\geq 1}$ ,  $|G| \in \mathbb{Z}_{\geq 1}$   $\therefore \frac{|G|}{p^m} \in \mathbb{Z}_{\geq 1}$ . 记  $k = \frac{|G|}{p^m}$

$\therefore k \in \mathbb{Z}_{\geq 1}$  且  $|G| = p^m \cdot k$ .

假设  $p \mid k$ , 则  $\because p \mid k$ ,  $k \in \mathbb{Z}_{\geq 1}$ ,  $p \in \mathbb{Z}_{\geq 2}$   $\therefore \frac{k}{p} \in \mathbb{Z}_{\geq 1}$

$\therefore \frac{p^m \cdot k}{p^{m+1}} \in \mathbb{Z}_{\geq 1}$   $\therefore \frac{|G|}{p^{m+1}} \in \mathbb{Z}_{\geq 1}$   $\therefore p^{m+1} \parallel |G|$  矛盾.  $\therefore p \nmid k$

$\therefore |G| = p^m \cdot k$ ,  $m \in \mathbb{Z}_{\geq 0}$ ,  $k \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid k$ .  $\square$

Lemma:  $G$  是有限群,  $p$  是素数, 记  $G$  的所有 Sylow  $p$ -子群组成的集合为  $\text{Syl}_p(G)$ , 记  $n_p = |\text{Syl}_p(G)|$ . 则有:

~~$n_p = |\text{Syl}_p(G)|$~~   $n_p = |\text{Syl}_p(G)| \in \mathbb{Z}_{\geq 1}$ .

Proof:  $\because G$  是有限群,  $p$  是素数  $\therefore |G| = p^m \cdot k$ , 其中  $m \in \mathbb{Z}_{\geq 0}$ ,

$k \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid k$ .  $\because m \in \mathbb{Z}_{\geq 0} \therefore p^m \in \mathbb{Z}_{\geq 1} \therefore \frac{|G|}{p^m} = k \in \mathbb{Z}_{\geq 1}$

$\therefore p^m \mid |G|$ .  $\because G$  是有限群,  $p$  是素数,  $m \in \mathbb{Z}_{\geq 0}$ ,  $p^m \mid |G|$

$\therefore G$  中一定有为阶为  $p^m$  的子群. 设  $H \leq G$  且  $|H| = p^m$

$\therefore p^m \parallel |G| \therefore H$  是  $G$  的 Sylow  $p$ -子群  $H \in \text{Syl}_p(G)$

$\therefore \text{Syl}_p(G) \neq \emptyset$ .  $\because G$  是有限群  $\therefore G$  的子集只有有限个.

$\therefore G$  的子群只有有限个  $\therefore G$  的 Sylow  $p$ -子群只有有限个.

$\therefore n_p = |\text{Syl}_p(G)| \in \mathbb{Z}_{\geq 1} \quad \square$

Lemma:  $G$  是有限群,  $p$  是素数, 对  $\forall G$  的  $p$ -子群  $H$ , 对  $\forall P \in \text{Syl}_p(G)$ ,  $\exists g \in G$ , s.t.  $H \leq gPg^{-1}$ .

Proof: 对  $\forall G$  的  $p$ -子群  $H$ , 对  $\forall P \in \text{Syl}_p(G)$ , 有:

$\because H$  是  $G$  的  $p$ -子群  $\therefore H \leq G$  且  $H$  是  $p$ -群.

记  $X = \{xPx^{-1} \mid x \in G\} \subseteq \text{Syl}_p(G)$

$\because P \in \text{Syl}_p(G) \therefore p \nmid (G:P)$  且  $(G:P) \in \mathbb{Z}_{\geq 1}$

$\because G$  是有限群,  $P \leq G$ ,  $p$  是素数,  $(G:P) \in \mathbb{Z}_{\geq 1}$ ,  $p \nmid (G:P)$

$X = \{xPx^{-1} \mid x \in G\} \therefore X$  是非空有限集合, 并且  $p \nmid |X|$

$\because G$  是有限群,  $P \leq G$ ,  $X = \{xPx^{-1} \mid x \in G\}$ ,  $H \leq G$

$\therefore$  定义群作用  $\alpha: H \times X \longrightarrow X$  则  $\alpha$  是群作用.

$$(g, xPx^{-1}) \mapsto (gx)P(x^{-1}g^{-1})$$

$\because H$  是 ~~有限~~ 有限群,  $H$  是  $p$ -群,  $X$  是非空有限集合,  $p \nmid |X|$

$$\therefore X^H \neq \emptyset. \quad \therefore \exists Q \in X^H \quad \therefore Q \in X$$

$$\text{对 } \forall h \in H, \therefore Q \in X^H \quad \therefore \alpha(h, Q) = Q \quad \therefore hQh^{-1} = Q$$

$$\therefore Q \in X \subseteq \text{Syl}_p(G) \quad \therefore Q \text{ 是 } G \text{ 的 Sylow } p\text{-子群}$$

$$\therefore h \in G \text{ 且 } hQh^{-1} = Q \quad \therefore h \in N_G(Q) \quad \therefore H \leq N_G(Q)$$

$$\therefore N_G(Q) \leq G \quad \therefore H \leq N_G(Q) \quad \therefore H \leq Q$$

$$\therefore Q \in X \quad \therefore \exists \lambda \in G, \text{ s.t. } Q = \lambda P \lambda^{-1}$$

$$\therefore \lambda \in G \text{ 且 } H \leq \lambda P \lambda^{-1} \quad \square$$

~~推论~~ 推论:  $G$  是有限群,  $p$  是素数, 对  $\forall G$  的  $p$ -子群  $H$ ,

$$\exists Q \in \text{Syl}_p(G), \text{ s.t. } H \leq Q.$$

proof: 对  $\forall G$  的  $p$ -子群  $H$ , 任取  $P \in \text{Syl}_p(G)$ .

$$\exists g \in G, \text{ s.t. } H \leq gPg^{-1}. \quad \because g \in G \quad \therefore gPg^{-1} \in \text{Syl}_p(G)$$

$$\therefore gPg^{-1} \in \text{Syl}_p(G) \text{ 且 } H \leq gPg^{-1} \quad \square$$

Lemma:  $G$  是有限群,  $p$  是素数, 对  $\forall P \in \text{Syl}_p(G)$ , 有:

$$\text{Syl}_p(G) = \{xPx^{-1} \mid x \in G\}$$

Proof: 右边  $\subseteq$  左边 已证. 对  $\forall R \in \text{Syl}_p(G)$ , 有:  $R$  是  $G$  的

Sylow  $p$ -子群  $\therefore R$  是  $G$  的  $p$ -子群

$\therefore R$  是  $G$  的  $p$ -子群,  $P \in \text{Syl}_p(G) \therefore \exists g \in G$ , s.t.  $R \leq gPg^{-1}$

$\therefore g \in G \therefore gPg^{-1} \in \text{Syl}_p(G) \therefore R \in \text{Syl}_p(G) \therefore |R| = |gPg^{-1}|$

$\therefore R$  是有限群,  $gPg^{-1}$  是有限群,  $R \leq gPg^{-1}$ ,  $|R| = |gPg^{-1}|$

$\therefore R = gPg^{-1} \therefore R \in \text{右边} \therefore \text{左边} \subseteq \text{右边} \therefore \text{左边} = \text{右边}. \square$